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RESEARCH ARTICLE

Hybrid Intelligence for Field-Scale Soil Analysis and Crop Advisory Using Embedded Sensors and Machine Learning

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ABSTRACT This work provides the design and development of a portable soil testing system utilizing artificial intelligence (AI) and machine learning (ML) capabilities for real-time soil analysis and crop suggestion. It uses a series of sensors to measure the essential soil parameters of pH, moisture, temperature, electrical conductivity, and macronutrients-nitrogen (N), phosphorus (P), and potassium (K). These sensor data are processed in a microcontroller-based system and analyzed by applying ML algorithms in order to assess the fertility and categorize the soil. A rule-based system integrated with a Random Forest Regressor provides dynamic, explainable crop recommendation using soil condition, environmental conditions, and past data trends. Users can view real-time sensor data in a web-based visualization, monitor history, and receive AI-driven crop advisory support through a web-based dashboard. This field-friendly device is rugged, low in cost, and oriented towards the requirements of smallholder producers. Field experiments using 500 soil samples also exhibit high conformity to a commercial soil tester, as the respective RMSEs are 2.26 mg/kg for nitrogen, 1.90 mg/kg for phosphorus, 1.97 mg/kg for potassium, and 0.60 for pH. The deployed ML model had an R^2 value of 0.978 in test data. These findings validate the system's accuracy, dependability, and potential to facilitate sustainable, data-based decision-making in precision agriculture.

INDEX TERMS Internet of Things (IoT), precision agriculture, soil analysis, artificial intelligence, machine learning, crop recommendation, smart agriculture, embedded systems, web application, precision farming, environmental monitoring.

I. INTRODUCTION

Soil quality is a critical determinant of agricultural productivity, crop health, and environmental sustainability.

However, conventional soil testing methods often require laboratory analysis, which can be expensive, with a single comprehensive analysis often exceeding 30-50 US dollars per sample, rendering it time-consuming and inaccessible to smallholder farmers especially in rural and developing regions [3], [6], [18].

These barriers delay essential decisions about fertilization and crop planning, leading to inefficient resource use and increased production costs. In countries like Thailand, where

many farmers operate on limited budgets and have low gross domestic product (GDP) per capita, especially in rural agricultural zones, access to professional soil testing services is even more constrained. As a result, farmers often apply fertilizers or select crops based on guesswork rather than data, which can harm both crop yield and long-term soil health. To address these challenges, the integration of artificial intelligence (AI), machine learning (ML), and IoT-based sensing systems has enabled the development of smart, portable, and cost-effective soil analysis tools [1], [2], [5]. These systems can measure key soil parameters—such as pH, moisture, temperature, electrical conductivity, and macronutrients (NPK: nitrogen, phosphorus, potassium)—on-site and in real time [4], [8], [19]. By minimizing the need for laboratory involvement,

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these devices offer a low-cost alternative that makes frequent testing feasible and financially accessible for farmers in low-income areas.

While staple crops are typically cultivated on an annual basis, the intra-seasonal dynamics of soil nutrient availability necessitate frequent monitoring. Factors such as precipitation events, irrigation schedules, and nutrient uptake rates at various phenological stages can cause significant fluctuations in soil composition. Consequently, routine analysis enables timely and targeted interventions, such as precision fertilization, to mitigate nutrient deficiencies or toxicities that can adversely affect crop yield. Thus, the proposed system is not limited to pre-sowing crop selection but also functions as an essential tool for dynamic resource management throughout the cultivation period.

This research proposes a portable AI-based soil testing device that combines multi-sensor integration, embedded computing, and machine learning algorithms to assess soil conditions and generate online crop advisory recommendations. The system includes a web platform that allows users to review real-time soil health reports and receive AI-guided recommendations for crop selection and nutrient management [10], [23], [24]. Designed to be compact, affordable, and easy to operate, the system empowers farmers to make data-driven decisions, while helping to reduce unnecessary inputs, optimize yield, and lower farming costs. This work supports the advancement of precision agriculture and aligns with Thailand's broader goals of promoting technological adoption and improving economic outcomes for smallholder farmers under Agriculture 4.0 initiatives [6], [20], [28].

II. RELATED WORK

The adoption of smart agriculture technologies has become essential in enhancing crop productivity, resource optimization, and sustainability—particularly in regions where access to traditional agricultural infrastructure is limited. Soil testing is a foundational step in precision farming, enabling informed decisions on fertilization, crop selection, and water management. However, conventional soil testing methods are costly, time-consuming, and often inaccessible to smallholder farmers, especially in developing countries. In response, research has increasingly focused on developing integrated systems that combine low-cost sensors, embedded computing, artificial intelligence (AI), and digital platforms to support real-time, location-specific soil diagnostics. This section provides a detailed review of the current state-of-the-art across five critical domains: portable soil testing systems, nutrient sensing technologies, AI and machine learning applications, digital advisory platforms, and the remaining research gaps that motivate this study.

A. PORTABLE SOIL TESTING TECHNOLOGIES

Recent technological advancements have led to the development of various portable soil testing solutions aimed at replacing traditional laboratory analysis with compact,

field-deployable alternatives. These devices are designed to provide real-time soil insights, often relying on micro-controller platforms, wireless communication protocols, and compact sensor arrays. The objective is to offer farmers immediate, actionable data that informs decisions on fertilization, irrigation, and crop planning.

Khan et al. [3] and Verma et al. [4] were among the first to propose IoT-enabled soil monitors capable of recording and transmitting basic parameters like pH and moisture. Although effective in reducing diagnostic delays, these systems were often limited in scope and lacked nutrient assessment or crop recommendation functionalities. Sharma et al. [30] enhanced this approach using a Raspberry Pi to support multiple sensor inputs, achieving a high correlation with laboratory reference measurements. Their system demonstrated a balance between accuracy and affordability, positioning it as a viable option for field deployment.

Other efforts have explored autonomy and power efficiency. Ragi et al. [11] introduced a novel approach involving wireless power transmission through soil to activate remote sensor nodes, reducing the need for frequent maintenance or battery replacement in off-grid locations. Kim et al. [29] proposed a lightweight colorimetric sensor integrated with an AI algorithm to assess soil pH levels with a reported 97% accuracy rate. These developments underscore the increasing feasibility of building compact, self-sufficient systems for soil testing. However, challenges remain in scaling these technologies and ensuring their effectiveness in variable environmental conditions.

B. SENSOR-BASED SOIL NUTRIENT ANALYSIS

One of the most critical elements of precision agriculture is accurate nutrient profiling, particularly for macronutrients like nitrogen (N), phosphorus (P), and potassium (K). The conventional process of sending soil samples to a laboratory for chemical analysis is not only slow and expensive but also unsuitable for routine monitoring. As a result, the development of low-cost, in-field nutrient sensors has become a significant research focus.

Mane et al. [6] and Babu et al. [5] introduced systems using electrochemical and optical sensors for real-time nutrient assessment. Their results demonstrated good performance under controlled conditions but typically required skilled interpretation or lacked connectivity to decision-support systems. Rajput and Kumar [31] tackled long-term deployment challenges by designing a continuous soil pH sensor with embedded AI-driven calibration, reducing maintenance requirements. Patel et al. [8] expanded functionality by integrating LoRaWAN technology for long-range, low-power communication—enabling centralized data logging and analysis in rural environments.

Although significant progress has been made, most systems remain stand-alone units, with limited capability to perform intelligent analysis or interface directly with farmers through intuitive platforms. The opportunity lies in developing nutrient sensing systems that are not only accurate

and reliable but also capable of autonomously translating data into usable recommendations.

C. AI AND MACHINE LEARNING IN AGRICULTURE

Artificial intelligence (AI) and machine learning (ML) have introduced new possibilities for interpreting complex agricultural datasets and delivering predictive insights. In the context of soil analysis, ML models have been trained to classify soil types, predict nutrient levels, and recommend optimal crops based on localized data inputs. These models can substantially reduce the decision-making burden on farmers and improve efficiency across the agricultural value chain.

Wang et al. [2] and Zhang et al. [19] demonstrated the application of supervised learning techniques such as decision trees and support vector machines (SVMs) for crop suitability analysis. Their models utilized multidimensional soil data, including nutrient concentration and moisture, to recommend appropriate planting strategies. Nguyen et al. [7] applied real-time classification algorithms for determining soil health status based on sensor data, illustrating the role of AI in dynamic field environments.

However, as Sophocleous et al. [17] observed, the accuracy of AI predictions is heavily influenced by sensor calibration, data quality, and environmental noise. Moreover, many AI applications in agriculture rely on cloud computing resources, which are often unavailable or unreliable in rural areas. There is thus a growing emphasis on developing lightweight, embedded AI models that can operate on edge devices such as microcontrollers—allowing decision-making to occur directly in the field without the need for internet connectivity.

D. ONLINE AND WEB-BASED ADVISORY TOOLS

The use of online platforms and mobile applications in agriculture has expanded rapidly, offering a bridge between complex agricultural analytics and end users. These platforms collect soil and climate data—either manually entered or captured via sensors—and use AI models to deliver real-time recommendations. Common features include fertilizer management guidance, irrigation scheduling, and crop selection tools.

Roy and Sharma [23] and Rao et al. [24] proposed digital platforms that use user-input soil parameters to generate crop recommendations. These systems were successful in improving farmer awareness and promoting sustainable practices. Nevertheless, a major limitation lies in the reliance on accurate manual input and the availability of stable internet access—both of which are constrained in many rural regions.

More recent systems attempt to address this by integrating real-time sensor streams, automatically syncing field-collected data with cloud-based dashboards. While this integration offers enhanced automation, it also introduces new complexities in terms of cost, data security, and usability. A hybrid system—capable of operating both online and

offline while maintaining intelligent advisory functions—would offer greater flexibility and resilience in diverse agricultural settings.

E. GAPS AND RESEARCH MOTIVATION

Despite encouraging developments across sensor technologies, machine learning applications, and decision-support platforms, current solutions often fall short of offering a fully integrated, intelligent, and field-deployable system. Many existing tools specialize in a single domain, failing to address the entire workflow from data acquisition to actionable recommendation.

This fragmentation is evident when examining current approaches. For example:

IoT-based monitoring systems, such as the one proposed by Maddikunta et al. [37], excel at real-time data collection. What they provide is a constant stream of raw data. What they lack, however, is the integrated analytical capability to translate that data into practical advice. Our system bridges this gap by incorporating a hybrid intelligence model that provides immediate, actionable crop recommendations, moving beyond simple monitoring to active decision support.

Software-centric advisory platforms, like the one by Sohng and Kumar [38], offer powerful machine learning models for crop recommendation. What they provide is high predictive accuracy. What they lack is a connection to real-time field conditions, as they rely on manual data entry. In contrast, our work provides a fully integrated hardware-software solution, ensuring that the advisory is based on live, in-situ soil measurements, which eliminates the potential for human error and data staleness.

The emergence of Edge AI, as reviewed by Ferrag et al. [39], addresses the need for offline operation. What it provides is on-device processing for specialized tasks. What it often lacks is a comprehensive soil advisory solution. Our system offers a more versatile approach with its dual-mode capability: providing essential rule-based advice offline while delivering more sophisticated, data-driven insights when connected—a flexibility crucial for real-world agricultural settings.

This fragmentation—where systems are either data-rich but insight-poor, intelligent but disconnected, or offline but functionally limited—creates a significant gap, particularly in resource-limited environments such as rural Thailand, where farmers operate under financial constraints and lack access to reliable digital infrastructure [6]. Systems designed for high-tech environments are often ill-suited to these conditions due to assumptions about technical literacy, continuous connectivity, and capital availability.

As such, there is a critical need for a low-cost, compact, and intelligent soil testing system that bridges these specific gaps. Such a system must be able to operate independently of external infrastructure, analyze data locally using AI, and provide intuitive guidance through a user-friendly interface. The research presented in this work seeks to address this need by introducing a unified platform that integrates real-time

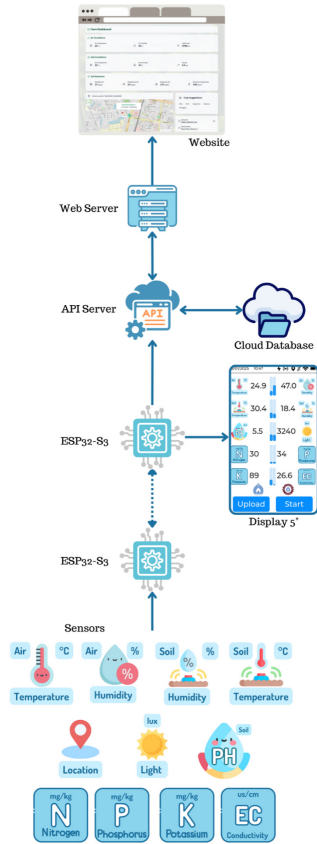


FIGURE 1. System architecture of the proposed smart soil analysis and crop advisory platform.

sensor monitoring, explainable AI-driven analysis, and both online and offline advisory capabilities—tailored specifically for the realities of smallholder farming.

III. METHODOLOGY

This section outlines the design and implementation of the proposed portable soil testing system, which integrates real-time sensing, embedded intelligence, and web-based crop recommendation. The methodology covers system architecture, sensor calibration, AI/ML algorithms, and software interface design. The goal is to provide a practical, low-cost, and user-friendly solution that enables smallholder farmers to assess soil health and receive actionable crop advice with minimal technical effort.

A. SYSTEM OVERVIEW

The system begins with a comprehensive sensor suite, which includes:

- Soil sensors for measuring pH, electrical conductivity (EC), soil temperature, soil moisture, and macronutrient levels—nitrogen (N), phosphorus (P), and potassium (K).
- Environmental sensors for capturing air temperature, air humidity, and ambient light intensity, which are important for assessing crop suitability and microclimate conditions.

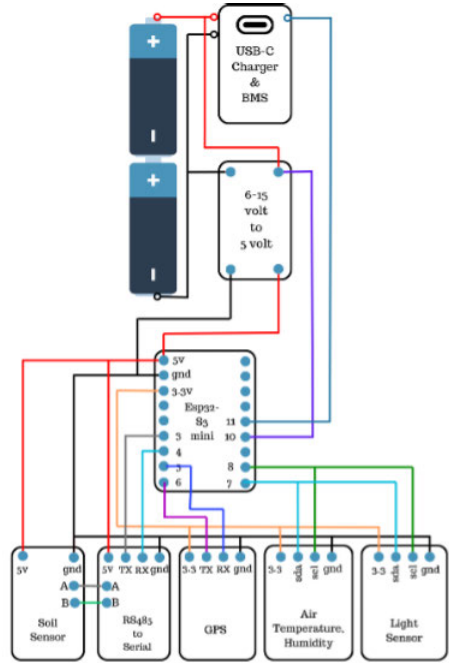


FIGURE 2. Block diagram illustrating the hardware components and their interconnections.

- A GPS module for geotagging each data point with precise location information.

All sensors interface with an ESP32-S3 microcontroller, which performs preliminary signal conditioning and data aggregation. The sensor values are transmitted to a secondary ESP32-S3 unit connected to a 5-inch touchscreen display, allowing users to interact with the system directly in the field. The screen displays real-time readings and includes interactive controls such as Upload and Start buttons for data transmission and device operation.

Collected data is sent to a centralized API server, which handles communication with a cloud database for storage and analytics. The data is then accessed through a web server, providing users with a dashboard where they can visualize soil health, monitor environmental metrics, and receive AI-generated crop recommendations.

This architecture supports real-time operation, offline interaction, and remote monitoring, making it well-suited for use in rural agricultural environments with limited infrastructure. The combination of soil, air, and light sensing offers a more holistic view of field conditions, supporting more accurate recommendations and farm management strategies.

B. HARDWARE ARCHITECTURE

The proposed system is structured around a modular, four-part hardware architecture optimized for portability, reliability, and real-time operation in low-resource agricultural environments. It consists of: (1) a sensor acquisition module, (2) a display and user interface module, (3) a power supply and battery management unit, and (4) a wireless communication subsystem. These components are distributed

across two ESP32-S3 microcontrollers that communicate wirelessly over a local Wi-Fi network created by a 4G dongle.

1) SENSOR ACQUISITION MODULE

The first ESP32-S3 microcontroller serves as the sensor node, responsible for interfacing with environmental sensors, preprocessing data, and hosting a RESTful API for data access. The module integrates the following components:

- **7-in-1 Soil Sensor:** Measures soil moisture, temperature, electrical conductivity (EC), pH, and macronutrients (nitrogen, phosphorus, potassium). It communicates via RS485 using the MODBUS RTU protocol and is connected to the ESP32 using an RS485-to-TTL serial converter (UART).
- **SHT35 Sensor (I²C):** Captures ambient air temperature and humidity.
- **BH1750 Light Sensor (I²C):** Measures ambient light intensity (lux), supporting decisions related to crop sunlight requirements.
- **GPS Module (UART):** Provides latitude and longitude for each measurement, allowing precise geotagging of soil data.

This ESP32-S3 runs an HTTP server, exposing REST API endpoints (e.g., /soil, /weather, /location) that return structured JSON data, which can be fetched by the display unit or remote services.

2) DISPLAY AND USER INTERFACE MODULE

The second microcontroller is the ESP32-8048S050C, an integrated ESP32-S3 board with a 5.0-inch 800 × 480 capacitive TFT display and a GT911 touch controller. This module serves as the system's user interface node, responsible for retrieving and displaying sensor data.

Key features include:

- Connects to the local Wi-Fi network created by the 4G dongle.
- Functions as a REST API client, periodically polling the sensor ESP32 via HTTP GET requests.
- Uses LVGL or equivalent GUI libraries to display real-time sensor values such as soil conditions, air metrics, and GPS coordinates.
- Touchscreen provides access to Start, Upload, Reset, and other system functions, making the platform highly user-friendly in field conditions.

By separating data acquisition and display, the system maintains modularity, improves performance, and supports scalability.

3) POWER SUPPLY AND BATTERY MANAGEMENT

The system is powered by two 18650 lithium-ion batteries connected in series, providing a nominal voltage of 7.4V and up to 8.4V when fully charged. This configuration enables several hours of continuous field operation with efficient power regulation and protection mechanisms.

Key components of the power subsystem include:

- **DC-DC Step-Down Converter (LM2596):** Regulates the 7.4V input from the battery pack to a stable 5V output, supplying power to the ESP32 microcontrollers, environmental sensors, and peripheral devices.
- **Lithium Battery Charging Module with Battery Management System (BMS):** Provides essential safety features, including protection against overcharging, over-discharging, overcurrent, and short circuits.
- **Physical Power Switch:** Allows manual control for powering the system on or off, thereby simplifying field use and extending operational life.
- **LED Indicators:** Includes a battery level indicator that visualizes charge status using voltage thresholds, and a system status LED that reflects device activity such as startup, data acquisition, upload, or idle mode.

This power configuration supports safe, reliable, and user-friendly energy management, with optional support for solar charging in off-grid agricultural environments.

4) WIRELESS COMMUNICATION SUBSYSTEM

The system's communication architecture is designed for field deployment by leveraging a cellular-to-Wi-Fi strategy. An integrated 4G USB dongle provides the primary internet uplink by connecting to available cellular networks. This dongle simultaneously functions as a portable Wi-Fi hotspot, creating a local wireless network that facilitates data exchange between the two ESP32-S3 microcontrollers. This approach makes the system independent of fixed Wi-Fi infrastructure and suitable for real-time cloud communication from any location with cellular service. Furthermore, to ensure operational robustness in areas with poor or no cellular coverage, the system is equipped with an offline data buffering capability. It can store multiple readings on local flash memory, allowing for a batch upload to the cloud once connectivity is restored.

The ESP32-S3 sensor node operates as a RESTful API server, exposing endpoints such as /data that deliver real-time sensor data in structured JSON format. Concurrently, the ESP32-8048S050C display node functions as a REST API client, periodically issuing HTTP GET requests to retrieve data and update the touchscreen interface accordingly.

This architecture enables seamless, local-area communication without requiring fixed network infrastructure, aside from the cellular data connection provided by the 4G dongle. The reliance on Wi-Fi not only supports physical modularity and remote deployment but also circumvents hardware limitations inherent to the microcontroller. Specifically, all UART ports on the ESP32-S3 sensor node are allocated to critical peripherals including the RS485-based soil sensor, GPS module, and optional debug interface. As such, Wi-Fi-based communication serves as an efficient, scalable, and non-intrusive alternative to wired serial communication.

C. SENSOR INTEGRATION AND CALIBRATION

Accurate sensor data is critical for reliable soil assessment and crop recommendation. This section outlines the process



FIGURE 3. Deployed sensor integration module for in-field soil data acquisition.

of integrating the sensor modules into the system and the calibration procedures used to ensure consistent and trustworthy measurements in field conditions.

1) SENSOR INTEGRATION

The sensor integration module is built around the ESP32-S3 microcontroller, which functions as the central data acquisition unit. It interfaces with multiple environmental and soil sensors using industry-standard protocols to enable accurate, real-time measurements in field conditions.

The 7-in-1 soil sensor communicates via RS485 (MOD-BUS RTU) through an RS485-to-TTL converter and provides readings for pH, moisture, temperature, electrical conductivity (EC), and macronutrients (N, P, K). Atmospheric data is captured by a SHT35 sensor (I²C) for air temperature and humidity, and a BH1750 light sensor (I²C) for ambient light intensity. A GPS module (UART) supplies geolocation data at 1 Hz for spatial tracking.

Sensor polling is handled via non-blocking routines in the ESP32 firmware. All data is formatted into JSON and exposed via a RESTful API for visualization and cloud upload. As shown in Figure 3, the sensor unit is mounted on a vertical pole with the probe embedded in soil and electronics housed in an upper enclosure for durability and ease of deployment in agricultural environments.

To ensure portability, modularity, and reliable performance, the Smart Soil Analyzer is enclosed in a compact unit integrating sensing, processing, communication, and power components. As illustrated in Figure 4, the ESP32-S3 microcontroller (1) acts as the central processor, interfacing with all sensors and managing data transmission via RESTful APIs. Power is supplied by a 7.4V dual-cell 18650 battery pack (3), regulated by a 5V step-down converter (2) and protected by a Battery Management System.

Sensor communication is handled via an RS485-to-TTL converter (4), while a GPS module (5) enables geotagging. A physical power switch (6) offers manual control, and

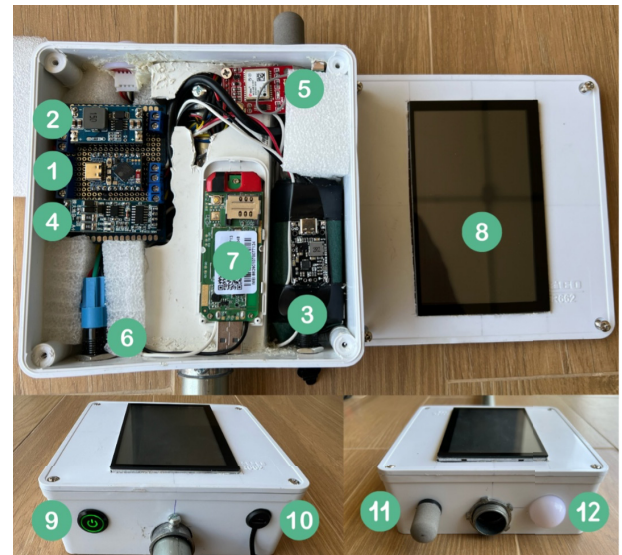


FIGURE 4. Physical assembly and layout of the Smart Soil Analyzer device.

a 4G Wi-Fi dongle (7) provides hotspot connectivity for ESP32 modules. The ESP32-8048S050C-based touchscreen display (8) offers a user-friendly interface with real-time data and control features. Additional components include a system status LED (9), USB-C charging port (10), and two environmental sensors—SHT35 (11) for air temperature and humidity, and BH1750 (12) for light intensity—mounted externally for ambient monitoring.

2) COMPARATIVE ANALYSIS WITH SOIL NUTRIENT INSTRUMENT

To evaluate the measurement consistency of the proposed soil sensing platform, a comparative analysis was conducted using the BK-Y6PC Soil Nutrient Tester, a commercially available device designed for in-field quantification of nitrogen (N), phosphorus (P), potassium (K), and soil pH. A total of ten soil samples were collected from various agricultural locations, homogenized thoroughly, and divided into two equal portions. One portion was analyzed using the integrated 7-in-1 RS485 soil sensor connected to the ESP32-S3 microcontroller, while the second portion was measured using the BK-Y6PC under equivalent environmental conditions.

All measurements were performed in a controlled indoor environment to minimize variations caused by ambient temperature, humidity, or light intensity. Sensor outputs from both systems were recorded and formatted for subsequent comparative analysis. This procedure was designed to establish baseline alignment between the proposed platform and a validated reference instrument through side-by-side evaluation.

The collected data were preserved for statistical regression analysis presented in Section IV. These results serve as the basis for evaluating the accuracy and reliability of the proposed sensor system for use in real-time, field-based agricultural applications.

D. AI AND MACHINE LEARNING FRAMEWORK

To intelligently support crop selection and soil improvement, the system implements a hybrid artificial intelligence framework that combines expert-driven rules with machine learning-based predictive modeling. This dual approach enables both interpretability and adaptability, allowing the system to deliver accurate, context-aware recommendations even in resource-constrained environments. The framework is divided into two core components: a rule-based AI expert system that evaluates soil conditions against agronomic thresholds, and a machine learning model trained on historical data to learn complex relationships between soil profiles and crop performance.

1) RULE-BASED AI EXPERT

The rule-based AI expert is designed to deliver explainable crop recommendations using a predefined knowledge base derived from agricultural best practices. This component is termed a 'Rule-Based AI' as it functions as an expert system, a classical branch of artificial intelligence that emulates the decision-making ability of a human expert by encoding domain knowledge into a set of 'if-then' rules. While it does not learn from new data in the manner of machine learning models, its capacity for automated reasoning based on a predefined knowledge base is a recognized characteristic of symbolic AI. This approach is especially useful for real-time, offline inference where cloud-based machine learning models may not be accessible. It evaluates sensor data against a table of acceptable soil parameter ranges for different crops and calculates a compatibility score for each.

a: SOIL FEATURE VECTOR DEFINITION

The input soil data is expressed as a vector:

$$\mathbf{X} = [x_1, x_2, \dots, x_n] \in \mathbb{R}^n \quad (1)$$

where x_i denotes the value of the i -th soil parameter. In this work: $x_1, x_2, x_3, x_4 = \text{pH, N, P, K}$

b: CROP SUITABILITY CONSTRAINTS

Each candidate crop $c \in C$ has an associated set of acceptable ranges for each parameter, defined as:

$$R_c = \left\{ \left[x_{i,\min}^{(c)}, x_{i,\max}^{(c)} \right] \mid \forall i \in \{1, \dots, n\} \right\} \quad (2)$$

where:

- $x_{i,\min}^{(c)}$ is the minimum acceptable value for parameter x_i for crop c
- $x_{i,\max}^{(c)}$ is the corresponding maximum value

These intervals are extracted from agronomic guidelines Table 1 and serve as the basis for rule evaluation.

c: SUITABILITY INDICATOR FUNCTION

To determine whether a specific parameter x_i satisfies the crop requirement, the system uses the following binary

TABLE 1. Acceptable Soil Parameter Ranges for Crops.

Crop Type	pH	Nitrogen(N)	Phosphorus(P)	Potassium(K)
Rice	5.5–6.5	80–150	15–25	60–100
Corn	5.5–7.0	90–130	15–30	80–120
Cassava	5.0–6.5	50–80	10–20	60–100
Sugarcane	5.5–7.5	100–150	20–40	100–200
Pineapple	4.5–6.0	40–70	10–25	80–150

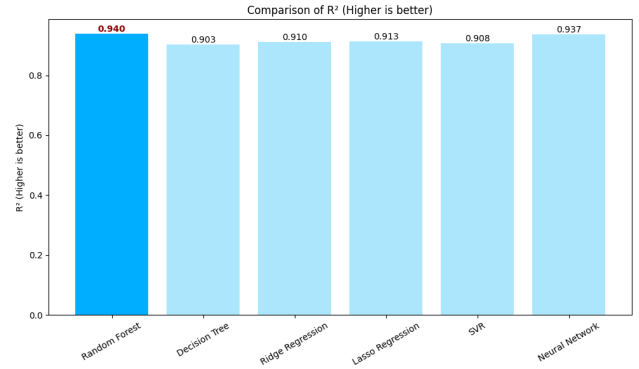


FIGURE 5. Average R² Scores of Machine Learning Models.

indicator function:

$$\delta_i^{(c)}(x_i) = \begin{cases} 1, & x_{i,\min}^{(c)} \leq x_i \leq x_{i,\max}^{(c)} \\ 0, & \text{otherwise} \end{cases} \quad (3)$$

This function returns 1 if parameter x_i is within the acceptable range for crop c , and 0 otherwise.

d: RULE-BASED MATCHING SCORE

For each crop, the system calculates a Match Ratio, representing the proportion of parameters that satisfy their respective constraints:

$$\text{MatchRatio}^{(c)} = \frac{1}{n} \sum_{i=1}^n \delta_i^{(c)}(x_i) \quad (4)$$

This scalar value ranges between 0 and 1. A MatchRatio of 1.0 indicates full compatibility between the soil condition and the crop's ideal growing conditions, while lower values reflect partial matches.

e: DECISION SUPPORT

All MatchRatios are evaluated for the candidate crop set C , and the results are passed to the hybrid recommendation engine. Crops with high MatchRatios are ranked higher and included in the final recommendation list. This rule-based AI system ensures that recommendations remain interpretable, fast, and robust even in the absence of connectivity or ML inference resources.

2) MACHINE LEARNING MODEL

To identify the optimal machine learning model for the system, we conducted a rigorous comparative evaluation of several candidate algorithms. The models were systematically assessed on their predictive accuracy using the

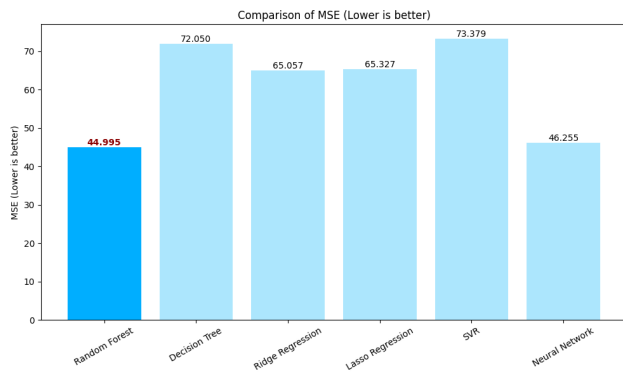


FIGURE 6. Average MSE Scores of Machine Learning Models.

coefficient of determination (R^2) and Mean Squared Error (MSE) as key performance metrics. To ensure statistical robustness, the evaluation was performed over 100 independent validation rounds.

The aggregated performance metrics are presented in Figure 5 and Figure 6. As shown in Figure 5, the Random Forest model achieved the highest average R^2 score of 0.940, indicating the best model fit. The Neural Network model also performed strongly with a close R^2 score of 0.937. Both significantly outperformed other models, such as the Decision Tree (0.903) and Support Vector Regression (SVR) (0.908).

Complementing these findings, Figure 6 illustrates that the Random Forest model yielded the lowest Mean Squared Error (MSE = 44.995), signifying the highest precision and lowest prediction error among all candidates. The Neural Network was the second-best performer (MSE = 46.255), while other models, particularly SVR (MSE = 73.379) and Decision Tree (MSE = 72.050), exhibited substantially higher errors.

Consequently, given its superior performance on both metrics—achieving the highest R^2 and the lowest MSE—the Random Forest Regressor (RFR) was selected for integration into our hybrid intelligence framework. Its exceptional predictive power, robustness against overfitting, and inherent ability to capture complex, non-linear relationships within the soil data make it the most suitable choice. Implemented in Python via the scikit-learn library, the RFR functions as the adaptive, data-driven component that complements the rule-based AI expert, thereby enhancing the overall precision and adaptability of the system's crop recommendations.

a: INPUT AND OUTPUT STRUCTURE

The model accepts a real-time soil feature vector defined as:

$$\mathbf{X} = [x_1, x_2, x_3, x_4] = [\text{pH}, N, P, K] \quad (5)$$

Each input $\mathbf{X}$ represents the current field readings from the soil sensor. The model outputs a Crop Suitability Score (MLScore) for each candidate crop, normalized between 0 and 1, where higher values indicate stronger compatibility with current soil conditions.

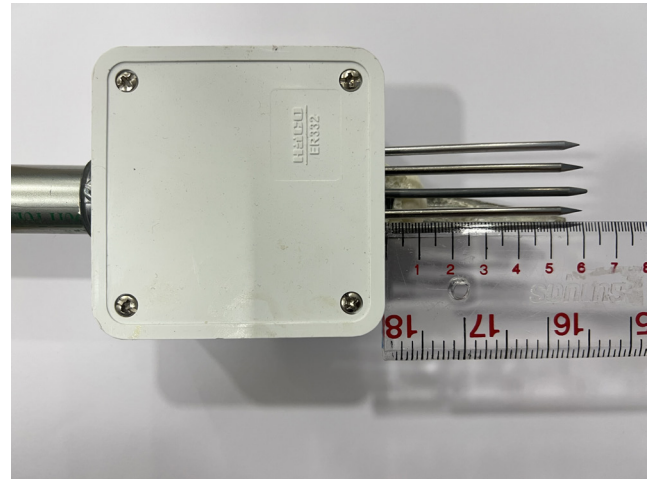


FIGURE 7. The length of the probe inserted into the soil.

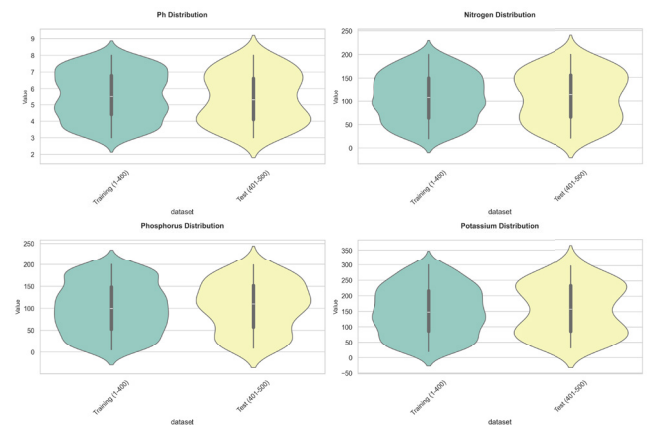


FIGURE 8. Violin plots illustrating the distribution of pH, nitrogen, phosphorus, and potassium in the training and test datasets.

b: DATASET COLLECTION AND CHARACTERISTICS

The dataset comprises 500 soil samples collected from agricultural areas in Kalasin Province, Thailand, a region known for cultivating a variety of key economic crops, including rice, sugarcane, pineapple, cassava, and corn. Soil samples were collected at a depth of 3-5 centimeters as shown in Figure 4, matching the operational depth of the developed sensor probe. The crop suitability labels used for training the model were derived from agronomic reference data published by Thailand's Ministry of Agriculture and Cooperatives. As shown in Table 1, This ensures that our model is grounded in nationally recognized knowledge and best practices, thereby enhancing the transparency and verifiability of our research process.

c: DATASET AND TRAINING PROCEDURE

The model was trained using a custom soil dataset consisting of 500 records. Each entry includes four soil chemical parameters—pH, nitrogen (N), phosphorus (P), and potassium (K)—along with corresponding AI crop suitability labels. The dataset was split into 400 training and 100 testing samples.

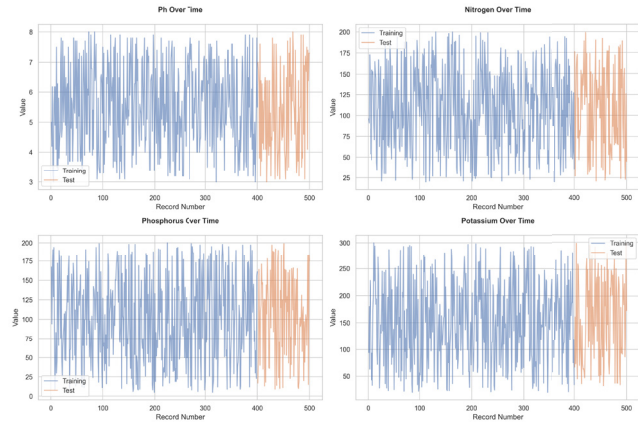


FIGURE 9. Time-series visualization of pH, nitrogen, phosphorus, and potassium across 500 records, split into training (1–400) and test (401–500) subsets.

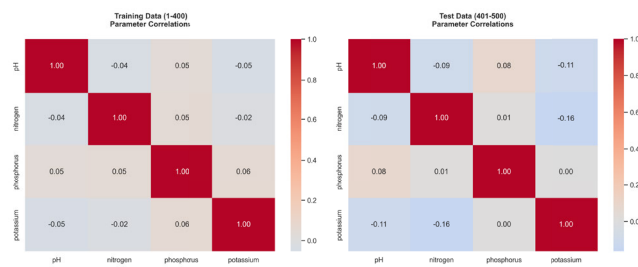


FIGURE 10. Correlation heatmaps comparing soil feature relationships in the training and test datasets.

To evaluate consistency across the training and test subsets, multiple visualizations were performed. Figure 8 shows the parameter distributions using violin plots, revealing well-balanced feature ranges across both subsets.

Time series trends for each parameter are shown in Figure 9, demonstrating temporal consistency and the absence of anomalies between training and testing segments.

Figure 10 presents Pearson correlation matrices for both training and test sets, indicating low multicollinearity and general consistency in feature relationships.

d: MODEL DEPLOYMENT AND INTEGRATION

Once trained, the model is serialized and deployed to a centralized server equipped with a RESTful API. During runtime, the ESP32-based sensor module transmits real-time soil data to the server using HTTP requests. The server executes the inference process and returns predicted MLScore values for each crop.

These scores are subsequently passed to the hybrid recommendation module, where they are fused with results from the rule-based AI engine. This architecture allows centralized model management, remote updates, and scalability without requiring firmware modifications to field-deployed hardware. It also supports future integration of additional crops or environmental parameters as new data becomes available.

3) HYBRID SCORING AND RECOMMENDATION STRATEGY

To generate reliable, explainable, and adaptive crop recommendations, the system fuses two intelligence layers: the rule-based AI expert and the machine learning model. This hybrid approach leverages the interpretability of expert-defined logic with the adaptability of learned patterns, enabling context-aware crop ranking across diverse field conditions.

a: FUSION METHODOLOGY

Each crop $c \in C$ is scored using a weighted combination of the rule-based Match Ratio and the machine learning-based MLScore, as shown below:

$$\text{HybridScore}^{(c)} = \alpha \cdot \text{MatchRatio}^{(c)} + (1 - \alpha) \cdot \text{MLScore}^{(c)} \quad (6)$$

where:

- $\text{MatchRatio}^{(c)} \in [0, 1]$ is computed by comparing real-time soil parameters with ideal crop-specific ranges.
- $\text{MLScore}^{(c)} \in [0, 1]$ is predicted by the Random Forest Regressor.
- $\alpha \in [0, 1]$ is a configurable weight (default: 0.6) determining the influence of each method.

This fusion ensures that each recommendation is both data-informed and agronomically valid.

b: HYBRID RECOMMENDATION ALGORITHM

The scoring procedure is formalized in Algorithm 1, which integrates rule-based and machine learning evaluations into a single hybrid recommendation pipeline:

To demonstrate the hybrid recommendation strategy, consider a real-time sensor input vector:

$$X = [\text{pH} = 6.2, N = 85, P = 18, K = 95]$$

The system evaluates this input against a predefined candidate crop set:

$$C = \{\text{Rice, Corn, Sugarcane, Cassava, Pineapple}\}$$

For each crop, the system calculates three metrics: the MatchRatio (based on expert-defined soil compatibility), the MLScore (predicted by the Random Forest model), and the HybridScore, which combines both metrics using a weighted average with $\alpha = 0.6$. The results are summarized in Table 2.

TABLE 2. Hybrid Crop Recommendation Output.

Crop	MatchRatio	MLScore	HybridScore
Rice	1.00	0.91	0.964
Corn	0.75	0.87	0.798
Sugarcane	0.50	0.79	0.658
Cassava	0.75	0.70	0.730
Pineapple	0.25	0.65	0.460

Based on the HybridScore values, the final ranked crop recommendations are:

[Rice, Corn, Cassava, Sugarcane, Pineapple]

Algorithm 1 RECOMMEND_CROPS($X, C, R, f_\theta, \alpha$)

```

1: Input:
2:    $X$  = Soil data vector [pH, N, P, K]
3:    $C$  = Set of candidate crops
4:    $R$  = Crop parameter range table
5:    $f_\theta$  = Trained ML model
6:    $\alpha$  = Rule-Based weight (e.g., 0.6)
7: Output: Ranked list of crops with scores
8: Initialize  $ResultTable \leftarrow \emptyset$ 
9: for each crop  $c \in C$  do
10:    $match\_count \leftarrow 0$ 
11:   for  $i \leftarrow 1$  to  $n$  do
12:      $x_i \leftarrow X[i]$ 
13:      $[x_{i,min}^{(c)}, x_{i,max}^{(c)}] \leftarrow R[c][i]$ 
14:     if  $x_i \in [x_{i,min}^{(c)}, x_{i,max}^{(c)}]$  then
15:        $\delta_i^{(c)} \leftarrow 1$ 
16:        $match\_count \leftarrow match\_count + 1$ 
17:     else
18:        $\delta_i^{(c)} \leftarrow 0$ 
19:     end if
20:   end for
21:    $MatchRatio^{(c)} \leftarrow match\_count / n$ 
22:    $MLScore^{(c)} \leftarrow f_\theta.predict\_proba(X)[c]$ 
23:    $HybridScore^{(c)} \leftarrow \alpha \cdot MatchRatio^{(c)} + (1 - \alpha) \cdot$ 
      $MLScore^{(c)}$ 
24:   Add  $(c, MatchRatio^{(c)}, MLScore^{(c)}, HybridScore^{(c)})$ 
     to  $ResultTable$ 
25: end for
26:  $RankedCrops \leftarrow \text{sort } ResultTable \text{ by } HybridScore^{(c)}$ 
     descending
27: return  $RankedCrops$ 

```

This hybrid scoring strategy integrates agronomic domain knowledge with machine learning-based inference to support precise, data-informed, and explainable crop decisions. The approach enhances flexibility and robustness, offering farmers practical guidance that is both grounded in expert knowledge and responsive to dynamic field data.

E. WEB-BASED DASHBOARD AND CLOUD INTEGRATION

The system improves accessibility, usability, and remote decision-making through a cloud-connected web-based dashboard. This interface acts as the primary point of interaction, enabling farmers and agronomists to visualize real-time soil data, track environmental conditions, and access AI-generated crop recommendations. By combining Internet of Things (IoT) devices with modern web technologies, the platform seamlessly connects low-cost field hardware to an intuitive digital environment. As a result, users with varying levels of technical expertise can easily leverage data-driven insights to enhance sustainable and efficient farming practices.

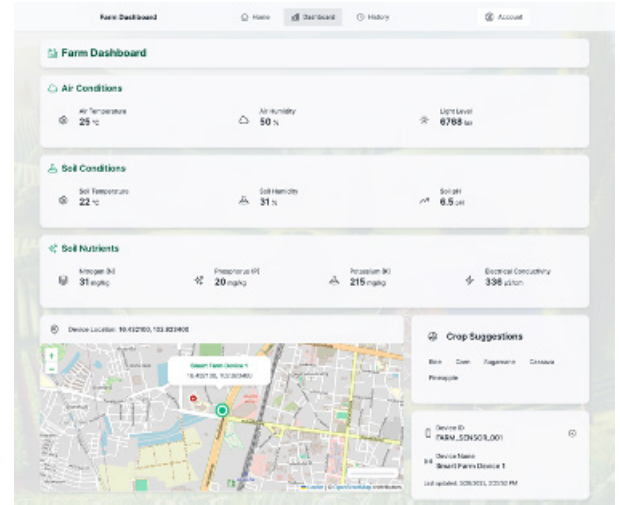


FIGURE 11. Web dashboard interface for real-time monitoring and crop recommendation visualization.

1) SYSTEM ARCHITECTURE

The web dashboard is built on a modular, full-stack architecture designed to ensure real-time responsiveness, scalability, and seamless integration with the field-deployed hardware. The system consists of four primary components:

- **ESP32 Sensor Node:** Captures real-time environmental data, including soil nutrients (N, P, K), pH, EC, temperature, moisture, light, humidity, and GPS coordinates. It communicates with the backend via Wi-Fi using HTTP POST requests.
- **ESP32 Display Node:** Acts as the user interface in the field, retrieving the latest crop recommendations from the backend and presenting them on a 5-inch touchscreen.
- **Backend Server (Node.js):** Implements RESTful API endpoints to receive data from ESP32 devices, forward sensor values to the AI modules, and store all readings in a database. It also provides responses to dashboard requests.
- **Database (MongoDB):** Stores all incoming sensor data, crop recommendations, timestamps, and GPS metadata in a structured NoSQL format for easy querying and historical analysis.
- **Frontend Interface (Next.js):** A responsive, React-based web application that allows users to monitor live data, view recommendation results, track historical entries, and explore field coverage via embedded maps.

The overall data flow is illustrated in Figure X, showing the communication paths between sensor nodes, backend logic, database, and frontend display components.

2) KEY FEATURES

The web-based dashboard provides an intuitive and interactive interface tailored for farmers, researchers, and agricultural extension workers. As shown in Figure 11, the platform delivers the following key functionalities:



FIGURE 12. Touchscreen dashboard on ESP32 for field-based soil data display.

- **Live Sensor Visualization:** Real-time display of soil and environmental parameters, including pH, nitrogen (N), phosphorus (P), potassium (K), electrical conductivity (EC), soil temperature, moisture, air temperature, humidity, and ambient light levels.
- **AI-Driven Crop Recommendation:** The top recommended crops, ranked by HybridScore, are visually presented in order of suitability, with indicators of rule-based and machine learning components for transparency.
- **GPS Mapping of Soil Data:** Each data entry is geo-tagged and displayed on an interactive map interface, enabling farmers to visualize soil conditions across spatial zones of their field.
- **Historical Data Log:** As illustrated in Figure 13, all sensor readings and recommendations are stored and can be accessed through filterable tables or graphs, allowing users to track soil changes over time and evaluate patterns.
- **Mobile-Friendly and Offline-Accessible Design:** Built using Next.js, the dashboard is optimized for both desktop and mobile devices, with progressive enhancement ensuring key functions remain usable even with intermittent connectivity.
- **User Interaction Controls:** Users can trigger real-time measurements, refresh recommendation results, or upload new datasets manually using the dashboard's control panel.

3) API AND DATA FLOW

The system leverages a modular communication architecture to bridge real-time data collection at the edge with intelligent cloud-based processing and user-facing services. This three-tiered system facilitates seamless interaction between the sensor node, display unit, and cloud backend through well-defined RESTful APIs. Each component communicates via standardized HTTP methods and JSON-formatted payloads, ensuring interoperability, scalability, and ease of integration with existing agricultural data platforms.

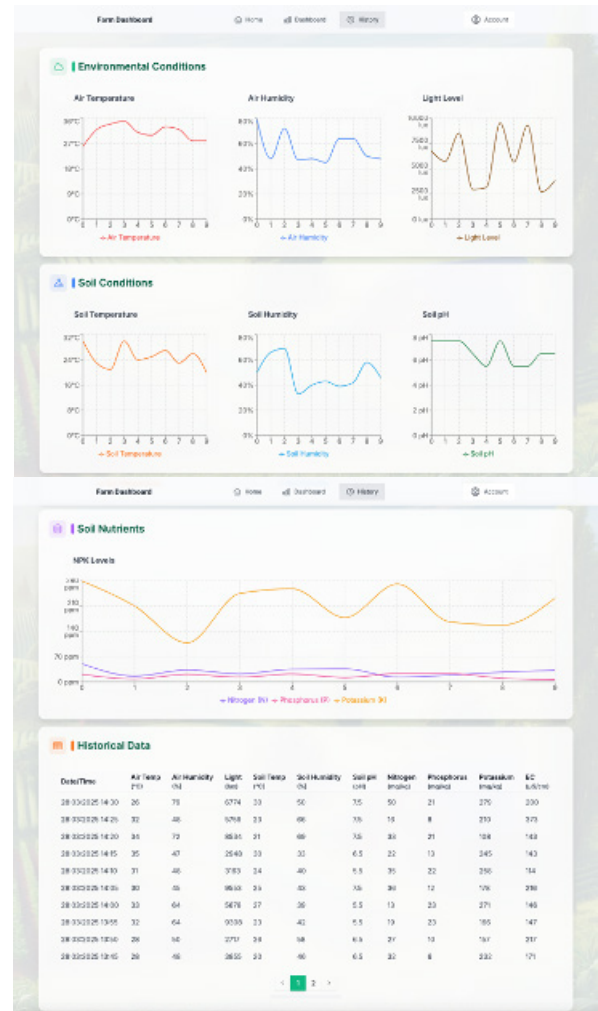


FIGURE 13. Historical data log showing previous sensor readings and recommendations.

a: LOCAL COMMUNICATION BETWEEN ESP32 MODULES

As illustrated in Fig. 14, the soil sensing system employs two ESP32-S3 microcontrollers to facilitate local communication: one acts as the sensor node (REST API server) and the other serves as the display module (API client). When the user interacts with the 5-inch touchscreen—typically by pressing the “Start” button—the display unit sends an HTTP POST request to initiate a new sensor reading. Once the measurement is completed, the latest environmental and soil data are retrieved via an HTTP GET request.

The primary RESTful endpoints exposed by the sensor node are:

- **POST /api/start** – Initiates a new sensor measurement cycle.
- **GET /api/data** – Retrieves the most recent soil and environmental metrics as a structured JSON object.

A sample API response from the sensor node is shown

```
{
  "date": "2025-01-10",
  "time": "15:25:30",
```

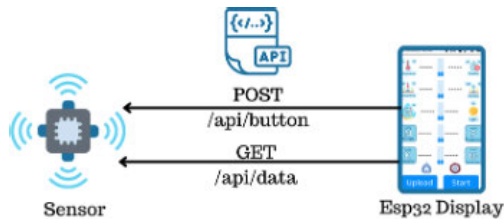


FIGURE 14. Local RESTful API communication between ESP32 modules for data acquisition and display.

```

``temperature``: 24.85,
``humidity``: 46.97,
``light_level``: 134,
``moisture``: 18.4,
``soil_temp``: 30.4,
``conductivity``: 26.6,
``pH``: 6.5,
``nitrogen``: 17,
``phosphorus``: 85,
``potassium``: 78,
``battery_level``: 100,
``charger_status``: 0,
``latitude``: 16.447543051,
``longitude``: 103.531181756,
``satellites``: 5
}

```

b: ESP32-TO-CLOUD BACKEND COMMUNICATION

After obtaining the latest soil and environmental readings from the ESP32 sensor node, the ESP32 display module forwards this data to a cloud backend using a standard HTTP POST request. The backend—implemented using Node.js—handles incoming data, activates the AI-driven crop recommendation system, and archives all records in a MongoDB database for long-term analysis.

The primary RESTful endpoint exposed by the backend is:

- POST /api/data/upload – Accepts a structured JSON payload containing sensor readings, timestamp, GPS coordinates, and device ID.

A typical JSON payload is as follows:

```

{
  ``device_id``: ``ESP32-Soil-001``,
  ``date``: ``2025-01-10``,
  ``time``: ``15:25:30``,
  ``pH``: 6.5,
  ``nitrogen``: 17,
  ``phosphorus``: 85,
  ``potassium``: 78,
  ``conductivity``: 26.6,
  ``moisture``: 18.4,
  ``soil_temp``: 30.4,
  ``light_level``: 134,
  ``humidity``: 46.97,
  ``temperature``: 24.85,

```

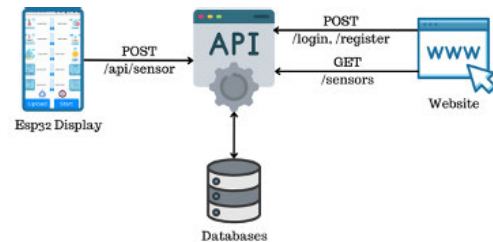


FIGURE 15. Cloud-based communication architecture for remote data storage and AI processing.

```

``battery_level``: 100,
``charger_status``: 0,
``gps``: {
  ``latitude``: 16.447543051,
  ``longitude``: 103.531181756,
  ``satellites``: 5
}

```

Upon receiving this payload, the backend performs the following actions:

- Validation: Confirms that all required fields are included and properly formatted.
- Storage: Saves the full dataset to a MongoDB collection, indexed by `device_id` and timestamp.
- AI Processing: Extracts key soil parameters (pH, N, P, K) and passes them to a hybrid AI engine (Random Forest Regressor + Rule-Based system) to generate crop suitability scores.
- Response: Sends back the top-ranked crop recommendations to the ESP32 display module for real-time user feedback.

c: FRONTEND-TO-BACKEND COMMUNICATION

The frontend dashboard, developed using the Next.js framework, interacts with the Node.js backend via a set of well-defined RESTful API endpoints. These endpoints enable the dashboard to retrieve the latest sensor readings, access AI-generated crop recommendations, and query historical records for analysis and decision-making support.

All API interactions are implemented using standard JavaScript libraries such as `fetch()` or `axios`, with responses returned in JSON format to maintain cross-platform and cross-browser compatibility.

The primary API endpoints are:

- GET /api/data/latest
Returns the most recent environmental sensor readings (e.g., pH, N, P, K, soil temperature, light level, etc.) for display in the live dashboard.
- GET /api/recommendations
Retrieves the latest crop recommendations produced by the hybrid AI engine, ranked by their respective suitability scores. Results are rendered in visual components such as cards or bar charts.

- GET /api/history

Provides access to all previously recorded datasets stored in MongoDB. Users may filter historical data by crop, timestamp, or GPS coordinates to support trend analysis and strategic planning.

The frontend issues asynchronous HTTP GET requests to these endpoints to ensure that the user interface remains responsive and continuously synchronized with backend logic. This architecture supports seamless data visualization and empowers users with interactive tools for monitoring soil conditions and evaluating recommendation outcomes.

F. SECURITY AND PRIVACY

The system ensures data integrity and confidentiality through a multi-layered security architecture. For data in transit, all API communications are secured with end-to-end encryption using HTTPS (TLS). Sensitive data payloads from the ESP32 are additionally encrypted with AES-256 prior to transmission. The web platform employs a robust authentication mechanism using JWT tokens, and user permissions are managed through Role-Based Access Control (RBAC). Furthermore, a comprehensive audit logging system is in place to track all security-relevant events, data access, and system configuration changes.

IV. RESULTS AND EVALUATION

This section presents the experimental validation and performance evaluation of the proposed Smart Soil Analyzer system. The results are organized into three key areas: (1) sensor accuracy and consistency through comparative testing against a commercial soil nutrient tester, (2) AI model performance including predictive accuracy and generalization capability, and (3) system usability and field deployment assessment under real-world agricultural conditions. These evaluations collectively demonstrate the effectiveness, reliability, and practical applicability of the system in supporting data-driven crop selection and soil health monitoring.

A. SENSOR ACCURACY AND RELIABILITY

To evaluate the accuracy and reliability of the proposed soil sensing system, a comparative experiment was conducted using the BK-Y6PC Soil Nutrient Tester, a field-grade commercial device recognized for its validated precision in measuring macronutrients and pH. Ten soil samples were collected from various agricultural sites, homogenized to ensure uniformity, and split into two equal parts. One portion was analyzed using the ESP32-based sensing platform with an integrated 7-in-1 RS485 soil sensor, while the second portion was tested using the BK-Y6PC device. All measurements were carried out under controlled indoor conditions to eliminate environmental variability.

The analysis focused on four key soil properties: nitrogen (N), phosphorus (P), potassium (K), and pH. Regression metrics were used to quantify the performance of the proposed system against the reference instrument. The results

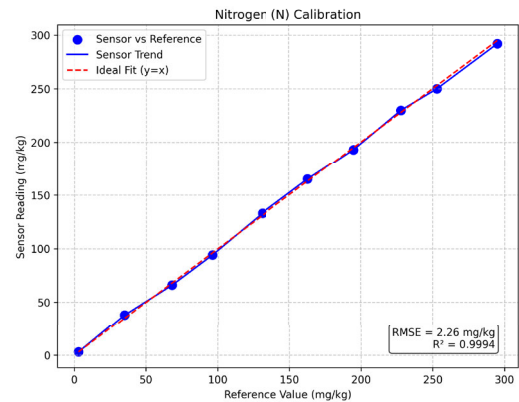


FIGURE 16. Calibration results for nitrogen (N) measurement: comparison between the proposed system and commercial reference.

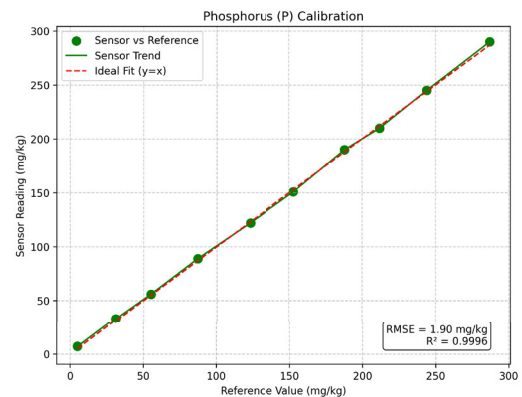


FIGURE 17. Calibration results for phosphorus (P) measurement: comparison between the proposed system and commercial reference.

demonstrate strong agreement between both systems. For nitrogen, the sensor achieved a Root Mean Squared Error (RMSE) of 2.26 mg/kg with a coefficient of determination (R^2) of 0.9994. Phosphorus measurements showed an RMSE of 1.90 mg/kg and an R^2 of 0.9996, while potassium exhibited an RMSE of 1.97 mg/kg with an R^2 of 0.9996. In the case of soil pH, the system yielded an RMSE of 0.60 and an R^2 of 0.9197, indicating reliable performance in acidity estimation.

The comparative calibration plots presented in Figures 16–19 demonstrate the strong correlation between the proposed system's measurements and the reference device. The high R^2 values (>0.99 for NPK and 0.92 for pH) combined with low RMSE values (2.26 mg/kg for N, 1.90 mg/kg for P, 1.97 mg/kg for K, and 0.60 for pH) validate the system's accuracy and reliability. These results confirm that the low-cost sensor integration provides measurement quality comparable to commercial-grade equipment, making it suitable for precision agriculture applications.

B. MACHINE LEARNING MODEL PERFORMANCE

The proposed crop recommendation system was evaluated using a Random Forest Regressor trained on a dataset comprising 500 soil profiles, each containing four input features: pH, nitrogen (N), phosphorus (P), and potassium

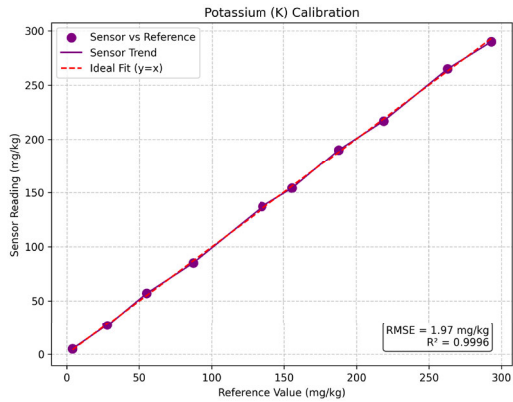


FIGURE 18. Calibration results for potassium (K) measurement: comparison between the proposed system and commercial reference.

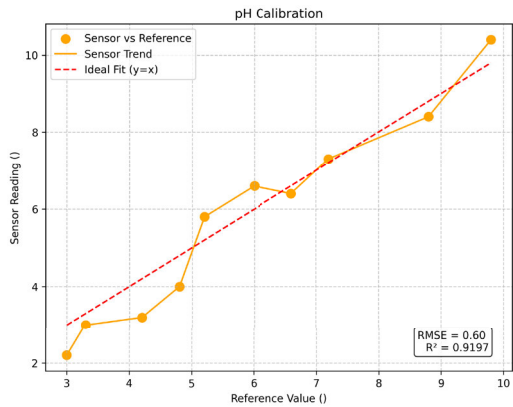


FIGURE 19. Calibration results for pH measurement: comparison between the proposed system and commercial reference.

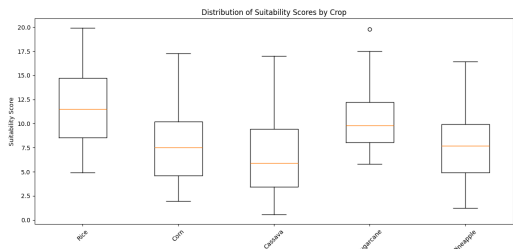


FIGURE 20. Distribution of predicted suitability scores for each crop.

(K). The dataset was partitioned into 400 training records and 100 test records. The model demonstrated exceptional predictive performance with a coefficient of determination (R^2) of 0.978 and a Root Mean Squared Error (RMSE) of 3.985, indicating an almost perfect match between predicted and actual suitability scores.

C. HYBRID SCORING AND CROP RECOMMENDATION

To provide explainable and effective crop recommendations, the system employs a hybrid scoring approach that combines rule-based matching with machine learning predictions. Each crop’s suitability is evaluated using both expert-defined parameter thresholds and the output of a Random Forest Regressor trained on historical suitability data.

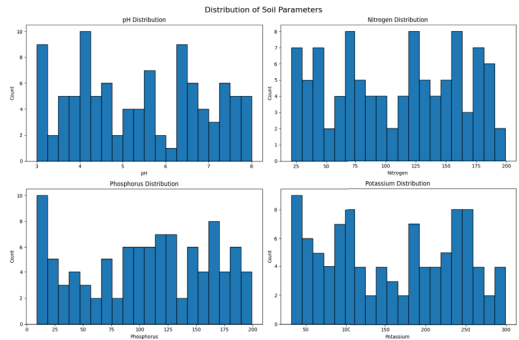


FIGURE 21. Distribution of soil parameters (pH, N, P, K) in the dataset.

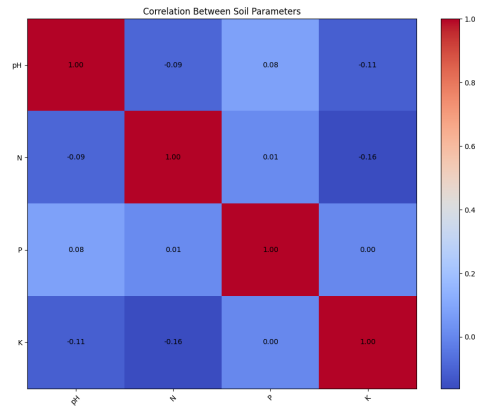


FIGURE 22. Correlation matrix of soil parameters in the dataset.

TABLE 3. Example Hybrid Scores for Crop Recommendation.

Crop	Match Ratio	ML Score	Hybrid Score
Rice	1.00	0.91	0.964
Corn	0.75	0.87	0.798
Cassava	0.75	0.70	0.730
Sugarcane	0.50	0.79	0.658
Pineapple	0.25	0.65	0.460

Given an input soil vector $X = [pH, N, P, K]$, the system computes:

- Rule-Based Match Ratio M_c , based on how many soil parameters fall within acceptable crop-specific ranges.
- ML Score S_c , predicted by the trained model based on the input features.
- Hybrid Score H_c , calculated as:

$$H_c = \alpha \cdot M_c + (1 - \alpha) \cdot S_c$$

where α is the weighting factor (e.g., $\alpha = 0.6$) emphasizing rule-based logic for interpretability.

This dual-layer evaluation ensures the model leverages both domain knowledge and data-driven learning. The crop with the highest hybrid score is ranked as the most suitable for the given soil condition.

For example, given a soil input $X = [pH = 6.2, N = 85, P = 18, K = 95]$, the system ranks the crops based on computed scores as shown in Table 3.

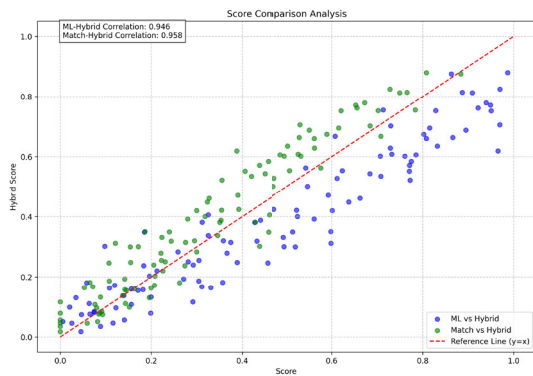


FIGURE 23. Comparison of hybrid score with ML score and match ratio for crop recommendation.

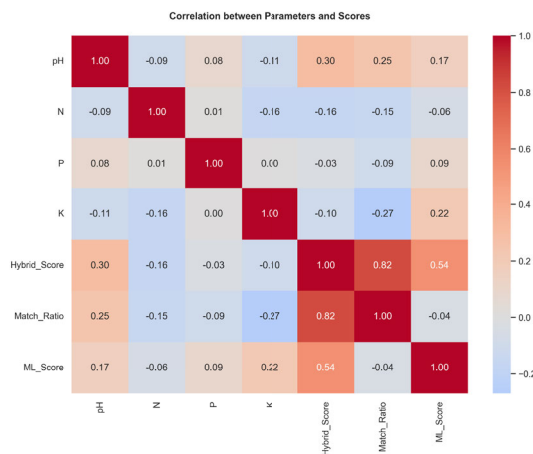


FIGURE 24. Correlation heatmap between soil parameters, ML scores, match ratio, and hybrid scores.

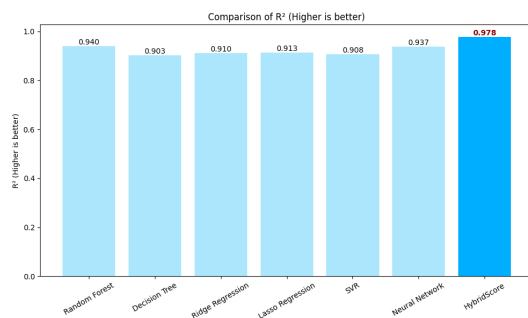


FIGURE 25. Comparison of average R^2 scores between hybrid and ML models.

To evaluate the reliability of the hybrid scoring method, we examined the relationship between the hybrid score and its two primary components: the rule-based Match Ratio and the AI-derived ML Score. As shown in Figure 23, the scatter plot compares each of these components with the final hybrid score. The Match Ratio (green) shows a notably higher correlation with the hybrid score (0.958) compared to the ML Score (blue), which has a correlation of 0.946. This confirms that with the chosen weighting parameter $\alpha = 0.6$, rule-based logic has a stronger influence on the final recommendation

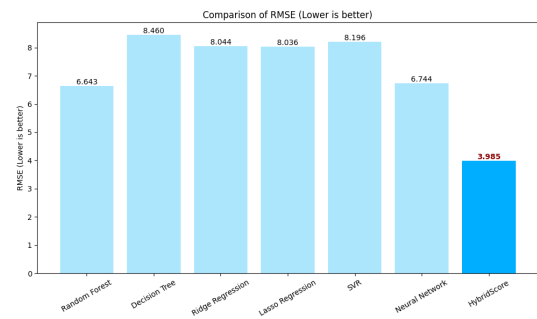


FIGURE 26. Comparison of average RMSE scores between hybrid and ML models.

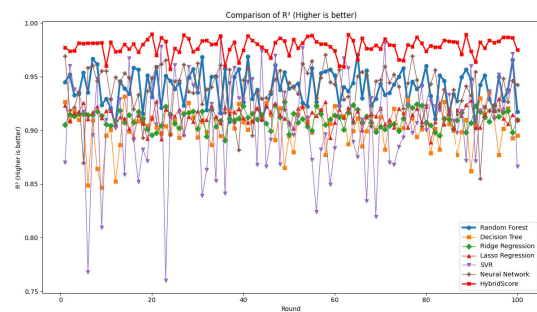


FIGURE 27. Line plot of R^2 scores over validation rounds for hybrid and ML models.

than the machine learning prediction. The alignment of points with the red reference line ($y = x$) further demonstrates the consistency and agreement between methods.

Complementing this, Figure 24 presents a correlation heatmap between soil parameters and the computed scoring metrics. The hybrid score exhibits moderate positive correlations with pH ($r = 0.30$), potassium ($r = 0.22$), and ML Score ($r = 0.54$), and a strong correlation with Match Ratio ($r = 0.82$). These patterns reflect the hybrid model's sensitivity to agronomic thresholds embedded in the rule-based approach, while still maintaining alignment with data-driven model outputs. The heatmap also confirms weak interdependence between individual soil parameters, supporting the assumption of low multicollinearity in the input space.

Together, these analyses reinforce the interpretability and consistency of the hybrid scoring system, affirming its suitability for real-world crop recommendation under heterogeneous soil conditions.

This interpretable scoring mechanism supports data-informed decisions while remaining intuitive to agronomists and farmers. By tuning the parameter α , the system can flexibly prioritize model predictions or expert rules based on use-case requirements.

This section presents a comprehensive and multi-faceted evaluation of the proposed system. The analysis is structured to validate three critical aspects: (1) the measurement fidelity of the integrated sensor suite, which forms the foundation of our data pipeline; (2) the predictive performance of various standalone machine learning models, establishing a

performance baseline; and (3) the effectiveness of our novel hybrid recommendation strategy, which represents the core contribution of this work.

The initial phase of our evaluation focused on the accuracy and reliability of the integrated soil sensor. Through rigorous comparative testing against a commercial-grade instrument, the sensor suite demonstrated high measurement fidelity. It achieved Root Mean Squared Error (RMSE) values below 2.3 mg/kg for macronutrients and an R^2 coefficient exceeding 0.91 for all measured parameters, including pH. This foundational result is crucial, as it confirms that the data fed into our intelligence layer is both accurate and reliable, thereby ensuring the integrity of subsequent analyses and recommendations.

The primary contribution of this research, however, lies in the development and validation of our hybrid scoring mechanism. This approach was designed to synergistically combine the interpretability of an expert-driven, rule-based system with the adaptive learning capabilities of a data-driven model. To quantify its effectiveness, we compared its performance against a suite of standalone machine learning models. The aggregated results, illustrated in Figure 25 and Figure 26, unequivocally demonstrate the superiority of the HybridScore. It achieved the highest average R^2 score of 0.978, indicating a near-perfect model fit to the data. This represents a significant improvement over the best-performing standalone model, the Neural Network ($R^2 = 0.937$). Concurrently, the HybridScore yielded the lowest average RMSE of 3.985, signifying a substantial reduction in prediction error compared to the Neural Network's RMSE of 6.744.

Furthermore, the stability of this performance was assessed over 100 validation rounds, as shown in Figure 27. The visualization clearly indicates that the HybridScore not only achieves a higher average R^2 but also exhibits significantly lower performance variance than the other models, which display more erratic fluctuations. This consistent, high-level performance underscores the robustness of our hybrid approach.

In conclusion, these results collectively validate that our hybrid strategy, which intelligently fuses agronomic domain knowledge with data-driven inference, produces crop suitability rankings that are demonstrably more accurate, reliable, and consistent than those generated by standalone machine learning methods. This confirms the system's enhanced applicability for precision agriculture and validates its potential for robust, trustworthy deployment in real-world, resource-constrained environments where both performance and explainability are paramount.

D. COMPUTATIONAL PERFORMANCE AND SCALABILITY

The system's architecture employs a client-server model, a strategic design choice that offloads computationally intensive machine learning tasks to a centralized cloud server. This approach preserves the low-cost, power-efficient nature of the field-deployed hardware, dedicating it solely

to data acquisition. This section evaluates the computational performance and scalability resulting from this architecture.

The computational efficiency of the selected machine learning models is a key factor in the system's performance. The Random Forest Regressor, chosen for its low complexity, results in a serialized model file of only 5.5 MB. This minimal footprint facilitates rapid loading and execution on the server.

System responsiveness was quantified by the end-to-end inference time, measured from data transmission by the field device to the reception of a recommendation. The evaluation demonstrated a highly efficient process, with an average inference time consistently under 1 second for all models, including the HybridScore. This low latency, directly attributable to the model's small footprint and the efficiency of the cloud infrastructure, is critical for delivering immediate, actionable feedback to users in real-world field conditions.

The cloud-based architecture ensures high scalability. System capacity is not constrained by the hardware limitations of the field device; rather, the cloud infrastructure can be scaled independently to accommodate a growing user base or more complex analytical models. This decoupling of field hardware and computational resources ensures the system's long-term viability and allows for future enhancements without impacting the low-cost nature of the deployed units. The analysis thus confirms the system's suitability for large-scale, real-world deployment where rapid and reliable feedback is essential.

V. CONCLUSION

This paper successfully demonstrates the design, implementation, and validation of a portable, hybrid intelligence system for field-scale soil analysis and crop advisory, tailored specifically for smallholder farmers in resource-constrained environments. Addressing the critical challenge of inaccessible and time-consuming laboratory-based soil testing, our work presents a fully integrated, end-to-end solution that bridges the gap from real-time data acquisition to actionable, on-the-spot recommendations.

The experimental validation confirms the system's high fidelity and robustness. Our integrated sensor suite achieved excellent measurement accuracy, with a strong correlation ($R^2 > 0.91$) and low error (RMSE < 2.3 mg/kg) when benchmarked against a commercial-grade instrument. The core innovation of this research—the novel hybrid scoring mechanism—was proven to be highly effective. By synergistically combining a data-driven Random Forest model with an explainable, rule-based expert system, the HybridScore achieved a superior R^2 of 0.978 and an RMSE of 3.985, significantly outperforming all tested standalone machine learning models. This key finding validates our hypothesis that the fusion of agronomic domain knowledge with data-driven inference leads to more accurate, reliable, and consistent crop recommendations.

Complemented by a user-friendly touchscreen interface for in-field operation and a cloud-connected web dashboard

for remote monitoring, the platform empowers farmers with accessible, data-driven insights. The system's dual-mode (online/offline) capability ensures practical usability in areas with intermittent connectivity. Ultimately, this work contributes a validated, low-cost, and intelligent tool that has the potential to advance precision agriculture, promote sustainable farming practices, and support the broader vision of a data-driven Agriculture 4.0.

VI. LIMITATIONS AND FUTURE WORK

While this study successfully establishes a robust proof-of-concept, we acknowledge several limitations that present clear avenues for future research and enhancement.

First, the machine learning model was trained on a dataset of 500 samples collected from a specific geographical region (Kalasin Province). Although representative of local conditions, this limits the model's generalizability to different soil types and agro-climatic zones. Future work will focus on a large-scale data collection effort, expanding the dataset to include samples from diverse agricultural regions across Thailand. This will help build a more robust and nationally applicable model, which will also incorporate mechanisms for detecting and handling anomalous sensor readings to improve real-world reliability.

Second, the current study lacks longitudinal field trials and a quantitative assessment of the system's real-world impact on crop yields, fertilizer usage, or its adoption rate by farmers. A crucial next step is to conduct long-term usability studies in collaboration with local farming communities to evaluate the system's practical benefits and gather feedback for iterative design improvements. These long-term studies will also be critical for assessing the sensor suite's stability and drift over extended periods in diverse field conditions and for developing a practical maintenance and calibration schedule.

Third, a detailed economic impact analysis, including return on investment (ROI) for farmers and a cost-benefit comparison with traditional methods, was not conducted. Similarly, this study did not include a formal environmental impact assessment of the device's lifecycle. Future interdisciplinary research should address these aspects to fully quantify the system's value proposition and ensure it aligns comprehensively with sustainable agriculture goals.

Fourth, while the hybrid model was designed for interpretability, future work could enhance transparency further by exploring advanced eXplainable AI (XAI) techniques, such as SHAP or LIME. These methods could provide deeper, feature-level insights into the model's predictions, increasing trust and adoption among end-users.

Fifth, the system's advisory capability is currently based on a static, real-time snapshot of soil conditions. To transition from reactive to proactive management, future work will explore the integration of Long Short-Term Memory (LSTM) networks to forecast future nutrient trends. Deploying such models on edge devices presents a challenge that may require the integration of specialized hardware accelerators.

Finally, the current sensor suite is limited to macronutrients (N, P, K). Future hardware iterations will aim to incorporate sensors for key micronutrients (e.g., Zinc, Boron, Iron) to provide a more holistic soil health assessment and further enhance the precision of our recommendations. We also recognize that while the system is designed to promote equitable access, future work should include a formal study of the broader social and ethical implications to ensure a responsible and inclusive technological rollout.

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